

⑨ Fault detection via HMM

Fault-detector: function $fd(t)$

taking System observations $o(t)$ as input and output estimate of whether a fault is present $o(t)$ either Seq. of observations or one observations

General 4 cases: 2 system states 2 detector outputs

Negative
FD=0

Positives
FD=1

$$TNR + FPR = 1$$

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$$S=0 \quad TNR := P_r(FD=0|S=0) \quad FPR := P_r(FD=1|S=0)$$

$$S=1 \quad FNR := P_r(FD=0|S=1) \quad TPR := P_r(FD=1|S=1)$$

$$\text{Accuracy} := P_r(FD=S) = TPR \cdot P_r(S=1) + TNR \cdot P_r(S=0)$$

Congestion detection example (using RTT)

- Detector can be $FD(t_i) = \mathbb{1}(RTT_i > \text{threshold})$

- Tradeoff thresh $\uparrow$ FPR $\downarrow$ & TPR $\downarrow$
ROC Curve show

$S=1$
 $S=0$



Use ROC Curve
to set
thresh

